

Alan Kay's Definition of OOP

- A system of independent objects that interact by passing messages.
- Objects encapsulate their state and behavior.
- Objects are dynamically defined and can evolve over time.

"Distributed, message-driven computation"

"if you have enough computers that can inter-communicate, you can define anything compatible".

"Very few in computing actually put in the effort to grok the implications of universal scalable systems of processes and instead have clung to very old and poorly scalable ways to program."

3) (Autonomous Objects) $\Rightarrow$ Objects act independently, running their own behavior based on received messages

4) (Late Binding) $\Rightarrow$ Objects determine behavior dynamically (promoting flexibility and scalability)

5) (Scalability) $\Rightarrow$ Alan Kay's model supports scalability and distributed computing, which is ideal for networked systems and parallel processing.

1) (Encapsulation) $\Rightarrow$ Each object maintains full control over its internal state and does not expose it directly.

2) (Message Passing) $\Rightarrow$ objects interact only through messages, avoiding direct methods calls.

- Encapsulation (1)
- Message Passing (2)
- Autonomous objects (3)
- Late Binding (4)
- Scalability (5)

